AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Chemistry for biologists

Living things are built from a small set of molecules, and most of their behaviour follows from the shapes of those molecules and the forces between them. This page covers water as a solvent, the carbohydrates and lipids that store energy, the bonds that join them, and the core practical for testing sugars and starch.

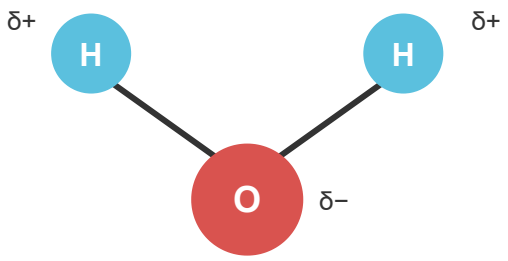
SPEC 1.1 Water: a polar solvent for transport

WHAT THIS SPEC POINT ASKS

Explain why water's dipole nature makes it a good solvent, and why that matters for transport.

Why water is a dipole

- In a water molecule, oxygen is more **electronegative** than hydrogen, so it pulls the shared electrons in each O–H covalent bond towards itself.
- Oxygen ends up slightly negative (**δ−**) and each hydrogen slightly positive (**δ+**). The molecule is bent, so the charges don't cancel: water is **polar** (a **dipole**).
- The δ+ hydrogen of one molecule is attracted to the δ− oxygen of a neighbour. This weak attraction is a **hydrogen bond**. Hydrogen bonds constantly break and re-form, which is why water is a liquid that flows.



δ− on oxygen, δ+ on each hydrogen

Why water is polar

1. Oxygen has a higher electronegativity than hydrogen.
2. The shared bonding electrons spend more time near the oxygen nucleus.

Electrons are pulled towards the atom that attracts them more strongly.

3. Oxygen becomes δ− and each hydrogen becomes δ+.
4. The molecule is bent, so the two bond dipoles add up instead of cancelling.

   A straight molecule with the same bonds would have no overall dipole.
5. The result is a permanent dipole, which lets water form hydrogen bonds with other polar molecules.

Why that makes water a good solvent

- Ions and polar molecules are attracted to water's charged regions:
  - δ− oxygens cluster around **positive ions** (e.g. Na^+); δ+ hydrogens cluster around **negative ions** (e.g. Cl^-).
  - The ions are pulled apart from each other, surrounded by water molecules, and **dissolve**.
- Polar molecules such as **glucose, amino acids** and **urea** form hydrogen bonds with water, so they dissolve too.
- Non-polar molecules (e.g. **lipids**) can't interact with water's charges, so they don't dissolve. Blood carries them another way (e.g. packaged as lipoproteins).
- Once dissolved, solutes can **react** (water is the medium for metabolic reactions) and can be **carried in solution**.

Link to transport

Where	Carried in solution
Blood plasma	glucose, amino acids, ions, hormones, urea, CO_2 (mostly as hydrogencarbonate)
Xylem (plants)	water and mineral ions
Phloem (plants)	sucrose and amino acids

MARK-SCHEME WORDING

- "oxygen (atom) is slightly negative and the hydrogens (atoms) are slightly positive". The charges must be **partial** (δ+ / δ−, "slight"), never full + and −.
- Why: "uneven charge distribution" / "oxygen pulls the electrons towards it".
- Diagram: two H joined to one O by covalent bonds, δ− on the O, δ+ on each H. A hydrogen bond runs from the δ+ H of one molecule to the δ− O of the next.
- Ions dissolve because water is "polar / dipolar": "sodium ions attracted to the oxygen", "chloride ions attracted to the hydrogens".

- Fatty acids are insoluble: "non-polar / hydrophobic tails" that "do not form hydrogen bonds" with water. "Too big" is ignored.
- "dipolar" is the preferred cloze word; "polar" is accepted.
- Solutes are "transported in solution" in plasma, xylem and phloem; water is the "medium for (metabolic) reactions".

Often asked as: draw a water molecule with partial charges, draw a hydrogen bond, draw water around Na^+, explain why something dissolves (or doesn't), describe a solubility-against-temperature graph.

COMMON MISTAKE

Saying water forms hydrogen bonds *with ions*. Ions are **attracted to the charged poles** of water molecules. Hydrogen bonds form between water and **polar molecules**, or between water molecules. And the oxygen and hydrogens are **atoms**, not "molecules": that slip loses the mark.

Full + and − charges, or charges on the wrong atoms. Hydrogen bonds drawn H to H or O to O.

"Glucose is too large to dissolve." It dissolves well because it is polar.

Calling a gradient a "rate" when time isn't on the x-axis.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Cohesion (water molecules hydrogen-bond to each other, so a column of water can be pulled up the xylem), **adhesion** (water sticks to vessel walls) and **high specific heat capacity** (blood and cells resist sudden temperature changes). Explain each one through hydrogen bonding.

Glucose is very soluble because its many **hydroxyl (–OH) groups** hydrogen-bond with water (WBI11 Jan 2020).

M_r of glucose: $C_6H_{12}O_6$ = (6 × 12) + (12 × 1) + (6 × 16) = **180**. You need to know the formula (WBI11 Jan 2020).

SPEC 1.2 Mono-, di- and polysaccharides

WHAT THIS SPEC POINT ASKS

(i) Know how monosaccharides, disaccharides and polysaccharides differ, including glycogen and starch (amylose and amylopectin). (ii) Relate their structures to their roles in providing and storing energy. (β-glucose and cellulose come later, in Topic 4.)

The three sizes of carbohydrate

Type	What it is	Examples	Soluble? Sweet?
Monosaccharide	single sugar unit (monomer), $(CH_2O)_n$	glucose, fructose, galactose (hexoses, $C_6H_{12}O_6$); ribose, deoxyribose (pentoses)	yes, yes
Disaccharide	two monosaccharides joined by a **glycosidic bond**	maltose, sucrose, lactose	yes, yes
Polysaccharide	many monosaccharides joined in a chain (polymer)	starch (amylose + amylopectin), glycogen	no, no

α-glucose

- Glucose is a **hexose** (6 carbons). Five carbons and an oxygen form the ring; carbon 6 sticks out as CH_2OH.
- In **α-glucose** the –OH on **carbon 1 points down**, on the same side as the –OH on carbon 4. (β-glucose, with the C1 –OH up, is Topic 4.)
- Glucose is the main **respiratory substrate**: small and **soluble**, so it is easily transported in blood and taken into cells, and its bonds store energy that respiration releases to make ATP.
- Disaccharides are quick energy too: they are digested into monosaccharides.

Storage polysaccharides: structure → function

	Amylose (starch)	**Amylopectin** (starch)	**Glycogen** (animals)
Glycosidic bonds	**1,4** only	1,4 and **1,6**	1,4 and **1,6**
Shape	unbranched chain that coils into a **helix**	**branched** (about every 20–30 glucose units)	**highly branched** (about every 8–12 units), more compact
What that gives	**compact**: lots of glucose in a small space	many **ends** → fast hydrolysis	most ends → **fastest** glucose release, suits animals' high metabolic rate
Where	starch grains in plant cells (amyloplasts/chloroplasts)		granules in **liver** and **muscle** cells

Properties that make all three good stores, with the reason for each:

- **Insoluble** → doesn't lower the **water potential** of the cell, so water isn't drawn in by osmosis; it also can't diffuse out of the cell.
- **Large** → can't cross the cell membrane, so the store stays where it is needed.
- **Compact** (coiled or branched) → a lot of energy stored in a small volume.
- **Branched** (amylopectin, glycogen) → many terminal glucose units, so many enzyme molecules can hydrolyse it at the same time, releasing glucose rapidly.
- Hydrolysis gives **α-glucose**, which can be respired straight away.

Why glycogen suits animal energy storage

1. Glycogen is a polymer of alpha-glucose, so it stores a lot of energy in a small volume.
2. It is insoluble and does not create an osmotic effect, so the cell does not take in extra water.

   Dissolved glucose molecules would each add to the solute concentration.
3. Its many branches give many non-reducing ends.
4. Enzymes can release glucose from every end at once, so glucose is released rapidly when needed.

   Rapid glucose release supports the high demand for respiration during exercise.

EXAM TIP

"Relate structure to function" questions want **pairs**: feature → so → consequence. "Branched, *so* there are many ends for enzymes to act on, *so* glucose is released quickly for respiration."

MARK-SCHEME WORDING

- Glycogen: "polysaccharide / polymer of α-glucose", "branched", "1–4 and 1–6 glycosidic bonds". Not "made of amylose and amylopectin".
- Link each feature to storage: "large, therefore insoluble / no osmotic effect" (and it can't diffuse out); "branched, so hydrolysis is faster"; "compact, so high energy density". "Easier to break down" is ignored.
- Amylose v amylopectin: **both** made of α-glucose, **both** have 1–4 bonds, both contain C, H and O only. Amylose: 1–4 only, unbranched, helical. Amylopectin: 1–4 and 1–6, branched.
- Starch is **insoluble**.

Often asked as: tick-grids (amylose v amylopectin v glycogen), compare-and-contrast, describe glycogen's structure, explain how glycogen suits storage, 6-mark on carbohydrate digestion, transport and storage.

COMMON MISTAKE

Stating "insoluble" on its own. Link it: large and insoluble, **so it has no osmotic effect** (it doesn't lower the cell's water potential) and can't diffuse out.

Thinking amylopectin has no 1–4 bonds: it has both 1–4 and 1–6.

Compare questions: give the **similarities** too, as paired statements ("both…", "amylose… but amylopectin…"). Points from two separate descriptions aren't pieced together.

Writing about properties when the question asks about structure. Glycogen isn't in plants, and starch isn't in animals.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Monosaccharide general formula: $C_nH_{2n}O_n$ (WBI11 Jan 2023).
- Liver glycogen is hydrolysed when blood glucose is too low; excess glucose is turned into fat (WBI11 Oct 2025).
- Undigested starch passes out in the faeces, so less glucose is absorbed. Starch itself is never absorbed (WBI11 Jan 2024).

SPEC 1.3 Core Practical 1: sugars and starch by colour standards

WHAT THIS SPEC POINT ASKS

Use Benedict's reagent and iodine solution, with colour standards, to estimate (semi-quantitatively) the concentration of reducing sugar and of starch.

Why Benedict's works

- A **reducing sugar** (all monosaccharides; maltose and lactose) can donate electrons. It reduces the blue **copper(II) sulfate** in Benedict's reagent to **copper(I) oxide**, a **brick-red precipitate**.
- More reducing sugar → more precipitate → colour runs **blue** → **green** → **yellow** → **orange** → **brick-red**.

CORE PRACTICAL

Core practical 1: method in brief

Reducing sugar (Benedict's)

1. Make a **dilution series** of glucose of known concentration (e.g. 0, 0.1, 0.2, 0.5, 1.0 mol dm^{-3}). For a **serial dilution**, dilute by the same factor each step (e.g. halve each time).

2. Add the **same volume** of Benedict's reagent (in **excess**) to the **same volume** of each solution.
3. Heat in a **water bath at the same temperature** (mark schemes use **80–95 °C**; some lab sheets say 60–80 °C, and either works if it is kept constant) for the **same time** (e.g. 5 minutes).
4. These tubes are your **colour standards**.
5. Test the unknown in exactly the same way and match its colour to the nearest standard.

Starch (iodine): make a starch dilution series and add the same volume of iodine solution to each. Colour runs **orange-brown** → **blue-black** as starch concentration rises. Compare the unknown with the standards. (No heating.)

Making it more accurate and reliable

- **Control variables**: volumes of sample and reagent, temperature, heating time, time before the colour is judged.
- Matching colours by eye is **subjective**, so the method is only *semi*-quantitative. Improve it with a **colorimeter**:
  1. Zero it with a blank (water), using a filter of contrasting colour.
  2. Measure the absorbance (or transmission) of each standard.
  3. Plot a **calibration curve** of absorbance against known concentration.
  4. Measure the unknown and **read its concentration off the curve**.
- Use **smaller concentration intervals** between standards for a more precise estimate.
- **Repeat** and calculate a mean; view the tubes against a **white tile/background**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Non-reducing sugars (e.g. sucrose) give a negative Benedict's result. To test for them: boil with **dilute hydrochloric acid** (hydrolyses the glycosidic bond), **neutralise** with sodium hydrogencarbonate, then repeat the Benedict's test. A colour change now shows a non-reducing sugar was present.

Unfamiliar colorimetric tests: use what each disaccharide is made of to predict results (e.g. a test for fructose detects sucrose but not maltose or lactose). If the calibration line doesn't pass through the origin, the test isn't sensitive enough to detect very low concentrations (WBI11 Jun 2023).

MARK-SCHEME WORDING

- "same volume of Benedict's", "same temperature and time of heating", "excess reagent", "compare with colour standards", "calibration curve", "colorimeter (removes subjectivity)".
- Calibration line not through the origin: "test is not sensitive enough" below a threshold concentration.

Often asked as: rarely examined step by step in WBI11 (2019–2026), but learn the method. More often: an unfamiliar colorimetric assay, or Benedict's as an MCQ distractor.

COMMON MISTAKE

Proposing Benedict's to detect sucrose without hydrolysing it first. Sucrose is a **non-reducing** sugar.

SPEC 1.4 Condensation and hydrolysis: glycosidic bonds

WHAT THIS SPEC POINT ASKS

Know how monosaccharides join by condensation reactions, forming glycosidic bonds, to make maltose, sucrose, lactose and polysaccharides, and how hydrolysis splits them.

Two mirror-image reactions

- **Condensation**: an –OH group on each sugar reacts; **one molecule of water is removed** and a **glycosidic bond** (C–O–C) forms between them.
- **Hydrolysis** (hydro = water, lysis = splitting): **water is added** across the glycosidic bond, breaking it. In living things it is catalysed by enzymes (amylase, maltase, sucrase, lactase), e.g. in digestion, or when liver and muscle cells break down glycogen.

Which sugars make which disaccharide

Disaccharide	Made from	Bond
Maltose	glucose + glucose	1,4-glycosidic
Sucrose	glucose + fructose	1,2-glycosidic
Lactose	glucose + galactose	1,4-glycosidic

Memory aid: **M**altose = a **M**atching pair; **S**ucrose has the fruit sugar; **L**actose is in mi**L**k (ga**L**actose).

- The bond is named by the carbons it joins: C1 of one sugar to C4 of the next is a **1,4** bond. Branches (amylopectin, glycogen) are **1,6** bonds.
- Polysaccharides form by **repeated** condensation. A chain of *n* glucose units releases *n* − 1 molecules of water.

EXAM TIP

To draw the condensation of two α-glucose: show the –OH on C1 of one and on C4 of the other, ring the H_2O being removed, then draw the product with **one O bridging C1 and C4**. Label it "1,4-glycosidic bond" and write **+ H_2O** as a product.

WORKED EXAMPLE

Worked example: M_r of a disaccharide

Glucose ($C_6H_{12}O_6$) has M_r 180. Each condensation removes one H_2O (M_r 18).

Maltose: 180 + 180 − 18 = **342** ($C_{12}H_{22}O_{11}$).

A chain of *n* glucose units: 180*n* − 18(*n* − 1).

MARK-SCHEME WORDING

- "condensation", "glycosidic (bond)", "hydrolysis" (**not** "hydration").
- Lactose = "galactose and glucose"; sucrose = glucose + fructose; maltose = glucose + glucose.
- Drawing: "glycosidic bond drawn correctly between C1 and C4" and water shown as a product. For hydrolysis, show H and OH added back to C1 and C4.
- Equation: $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + \mathbf{H_2O}$. The water is the part most often left out.
- Enzyme specificity: "the active site (of amylase) is complementary to the 1–4 glycosidic bond".

Often asked as: draw the condensation of two monosaccharides, write the equation, calculate an M_r, name the products of an enzyme acting on a polysaccharide.

COMMON MISTAKE

Swapping condensation and hydrolysis. Writing "lactase" (the enzyme) when you mean "lactose" (the sugar). Pairing the wrong monomers: lactose is glucose + **galactose**, not fructose.

In drawings, forgetting the water or losing atoms: every ring carbon keeps its H and OH groups.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Amylase hydrolyses only **1–4** bonds, so digesting amylopectin leaves small branched fragments that still contain the 1–6 bonds (WBI11 Jun 2024).
- Unfamiliar disaccharides appear. Work out the bond name from the carbon numbers drawn (WBI11 Jan 2022).

SPEC 1.5 Triglycerides: ester bonds, saturated vs unsaturated

WHAT THIS SPEC POINT ASKS

(i) Know how a triglyceride is made by condensation between glycerol and three fatty acids, forming ester bonds. (ii) Know how saturated and unsaturated lipids differ.

Building a triglyceride

- **Glycerol** is an alcohol with **three –OH groups**. A **fatty acid** is a –COOH (carboxyl) group attached to a hydrocarbon chain (written R–COOH).
- The –OH of glycerol reacts with the –COOH of a fatty acid in a **condensation** reaction: one H_2O is removed and an **ester bond** (–O–CO–) forms.
- One glycerol + three fatty acids → one triglyceride + **three water molecules**, with **three ester bonds**.
- **Hydrolysis** (catalysed by **lipase**) reverses this.

Saturated vs unsaturated

	Saturated	Unsaturated
C=C double bonds in the hydrocarbon chain	**none**: every carbon carries the maximum number of H atoms	**one** (mono-) or **more** (poly-unsaturated)
Chain shape	straight	**kinked** at each C=C
Packing	molecules pack closely	can't pack closely
At room temperature	usually **solid** (fats); mostly animal	usually **liquid** (oils); mostly plant
Health link	raise LDL and blood cholesterol → CVD risk (1.18)	tend to lower LDL

Why saturated fats are solid at room temperature

1. A saturated fatty acid tail has only single bonds, so it is straight.
2. Straight tails lie close together and their van der Waals forces add up across the whole length.
3. A lot of energy is needed to pull the tails apart.

More energy is needed to overcome the forces, so the melting point is higher.

4 At room temperature the fat stays solid.

Why triglycerides make a good energy store

- Many **C–H bonds** → about **twice the energy per gram** of carbohydrate.
- **Insoluble** (non-polar, hydrophobic) → no effect on water potential.
- Other roles: thermal **insulation**, **protection** of organs, buoyancy.

EXAM TIP

Melting-point data questions: melting point **rises with chain length** in saturated fatty acids and **falls as the number of C=C bonds rises**. If you say "positive correlation", **name both variables** or the mark isn't given.

MARK-SCHEME WORDING

- "one glycerol and three fatty acids", "condensation reaction", "ester bond" (esterification accepted), "catalysed by enzymes".
- Unsaturated: "at least one fatty acid has a C=C double bond"; fewer H atoms; the chain is "kinked".
- Saturated: say "no **C=C** double bonds". Plain "no double bonds" is rejected, because the C=O in –COOH is a double bond.
- Compare: **both** contain C, H and O, a hydrocarbon chain and a –COOH group. Ester bonds are not a feature of a fatty acid.
- Triglyceride v phospholipid: both have glycerol, fatty acids and ester bonds; a triglyceride has 3 fatty acids, a phospholipid has 2 plus a phosphate group.

Often asked as: describe how a triglyceride forms, draw one from given shapes (three fatty acids, three bonds, 4 bonds per carbon), compare saturated and unsaturated, interpret melting-point graphs.

COMMON MISTAKE

Calling a triglyceride a polymer: it is glycerol + 3 fatty acids, not a chain of repeating monomers. The bond is an **ester** bond (not glycosidic, not peptide), and **three** water molecules are released, not one.

Writing "glyceride + 3 fatty acids" or "3 glycerides". Drawing a saturated chain when an unsaturated one was asked for.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **C : H ratio**: a saturated fatty acid has more H, so a **lower** C : H ratio than an unsaturated one of the same length (WBI11 Jan 2026).
- Polyunsaturated fats (most C=C bonds) carry the lowest CVD risk (WBI11 Jun 2019).

Quick checks

QUICK CHECK

Which property of water is mainly responsible for the cooling effect of sweating?

- High specific heat capacity
- High latent heat of vaporisation
- High density at 4 °C
- Universal solvent

Sweat evaporates, and evaporation needs a large amount of energy to break hydrogen bonds. The energy comes from the skin, so the skin cools. Specific heat capacity describes temperature change inside a liquid, so it is the commonest wrong answer.

QUICK CHECK

A disaccharide is formed from two alpha-glucose molecules. What is released when the glycosidic bond forms?

- Carbon dioxide
- Oxygen
- Water
- A hydrogen molecule

Condensation joins two monosaccharides by combining an –OH group with an –H, and the two form water. Hydrolysis is the reverse. Students often think a bond forms by losing hydrogen gas, which is incorrect.

QUICK CHECK

Why is glycogen a better energy store in animal muscle than a single glucose molecule?

- It has a higher solubility in water
- It releases glucose more slowly
- It is insoluble and has many ends for rapid glucose release
- It contains more oxygen than glucose

Glycogen is insoluble, so it does not create a large osmotic effect, and its branches give many terminal glucose units for enzymes to act on at once. A common wrong answer is that glycogen is more soluble; it is actually the opposite.

QUICK CHECK

Which change would most likely lower the melting point of a fat?

- Making the fatty acid tails longer and fully saturated
- Introducing carbon–carbon double bonds into the fatty acid tails
- Joining the fatty acids to a larger glycerol molecule
- Increasing the number of hydrogen atoms in the tails

Double bonds create kinks that stop the tails packing together, which weakens the van der Waals forces and lowers the melting point. Longer saturated tails would pack more tightly and raise the melting point. Adding hydrogen to a tail is saturation, which raises the melting point.